Teaching Coastal Hazards Through Story And Field Work: Tsunami Science As A Native Scientist

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The Ohio State University College of Arts and Sciences

Roadmap to Curriculum

Observe with me. Explore with me. Interpret with me. Reflect together.

Story & Cultural Context

Introduce geology & geophysics through storytelling, community knowledge, and cultural relationships with the landscape.

Observation & Inquiry

Students observe coastal environments and ask questions about patterns, processes, and environmental change.

Classroom Exploration

Geoscience concepts such as tsunami generation, coastal morphology, and sediment transport are explored through collaborative activities and hazard scenarios.

Field Investigation

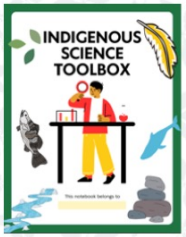
Students connect classroom learning to real-world landscapes through field observation and place-based exploration.

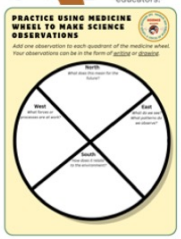
Reflection & Knowledge Sharing

Students interpret observations and reflect on how scientific knowledge connects to community resilience and environmental stewardship.

Curriculum Development & Future

Insights from classroom and field experiences inform the continued development of the Indigenous Science Toolbox, expanding learning resources and curriculum activities for future students and educators.





Scan Me

References

(Marshall et al., 2012), (Bang & Medin, 2010), (Bang et al., 2013), (Cajete, 2000), (Akerhead & Michell, 2011), (Semken, 2005), (Semken & Freeman, 2008), (Kimmerer, 2013), (Smith, 2012), (Buck & McKenzie, 2015)

Acknowledgements

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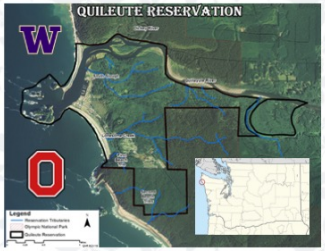
In honor of Mary Milwee

The Medicine Wheel Pedagogy

"The Medicine Wheel provides a systems-based pedagogy for connecting observation, culture, and geoscience in place-based hazard education."

Quileute Coast & Cascadia



Indigenous and Western Scientific Frameworks

Medicine Wheel Quadrant	Scientific Process	Learning Meaning
East – Observation *(Vision • Beginnings)*	**Observe & Ask Questions**	Curiosity begins with noticing patterns in the natural world and asking questions about what we see.
South – Connection *(Relationships • Understanding)*	**Research & Form Hypotheses**	Learners connect observations with prior knowledge, community knowledge, and scientific ideas.
West – Investigation *(Action • Experience)*	**Experiment & Collect Data**	Students test ideas through hands-on exploration, field observation, and data collection.
North – Reflection *(Wisdom • Insight)*	**Analyze & Interpret Results**	Data and observations are interpreted to understand the Earth processes shaping the landscape.
Center – Knowledge Sharing *(Balance • Community)*	**Communicate & Apply Knowledge**	Scientific knowledge is shared and applied to support community understanding and resilience.

CCASE Programs

Quileute CASE

"Indigenous science begins with observation of the natural world and relationships within it."
— Gregory Cajete (Tewa)

Lakota Concept	Meaning	Connection to Science
Mitákuye Oyás'iŋ	"All my relations" – everything is interconnected	Systems thinking in Earth science
Wóksape	Wisdom gained through experience	Long-term environmental observation
Wóohola	Respect	Ethical stewardship of land and water
Wólakota	Balance and harmony	Sustainable environmental relationships

Learning Together: Questions About Indigenous Science

(Leave a sticky note!)